

TCM Sheet 07

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Part a

The equation of state is

$$P = \frac{1}{\beta} \cdot \rho + \frac{1}{\beta} B_2 \cdot \rho^2 + \frac{1}{\beta} B_3 \cdot \rho^3 + \mathcal{O}(\rho^4) \quad (1)$$

$$\stackrel{\rho \ll 1}{\approx} \frac{1}{\beta} \cdot \rho + \frac{1}{\beta} B_2 \cdot \rho^2 \quad (2)$$

where the first term describes the ideal gas pressure and the second virial coefficient B_2 in a LJ system is given by

$$B_2 = -\frac{1}{2} \int_{\mathbb{R}^3} \left(e^{-\beta U(\mathbf{r})} - 1 \right) d^3\mathbf{r} \quad (3)$$

$$= -2\pi \int_0^\infty \left(\exp \left[-4\beta\epsilon \left(\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right) \right] - 1 \right) r^2 dr. \quad (4)$$

We define reduced (dimensionless) variables:

$$x = \frac{r}{\sigma} \quad (5)$$

$$\epsilon^* = \beta\epsilon \quad (6)$$

$$B_2^* = \frac{B_2}{\sigma^3}, \quad (7)$$

which, after some reformulations, result in the reduced form of the second virial coefficient

$$B_2^*(\epsilon^*) = -2\pi \int_0^\infty \left(\exp \left[-4\epsilon^* (x^{-12} - x^{-6}) \right] - 1 \right) x^2 dx. \quad (8)$$

Here, we plug in the given numerical values for ϵ^* in the range $[0, 1]$. This yields fig. 1, where $\epsilon^* = 0.5 \Rightarrow B_2^* = -1.31$ corresponding to argon (and also numerically given on the sheet) is drawn in as well.

In fig. 1, we can see that for small ϵ^* and thus weak particle interaction compared to the thermal fluctuations, the second virial coefficient B_2^* is larger than 0. This means that B_2 , the first order correction of the pressure to the ideal gas, will take an effectively repulsive form.

For larger ϵ^* which describe particle interactions roughly in the order of magnitude of the thermal fluctuations $B_2^* < 0$, thus attractive interaction dominates.

At $\epsilon^* \approx 0.3$, there seems to be an equilibrium between the attractive and repulsive corrections to the pressure since $B_2^* = 0$.

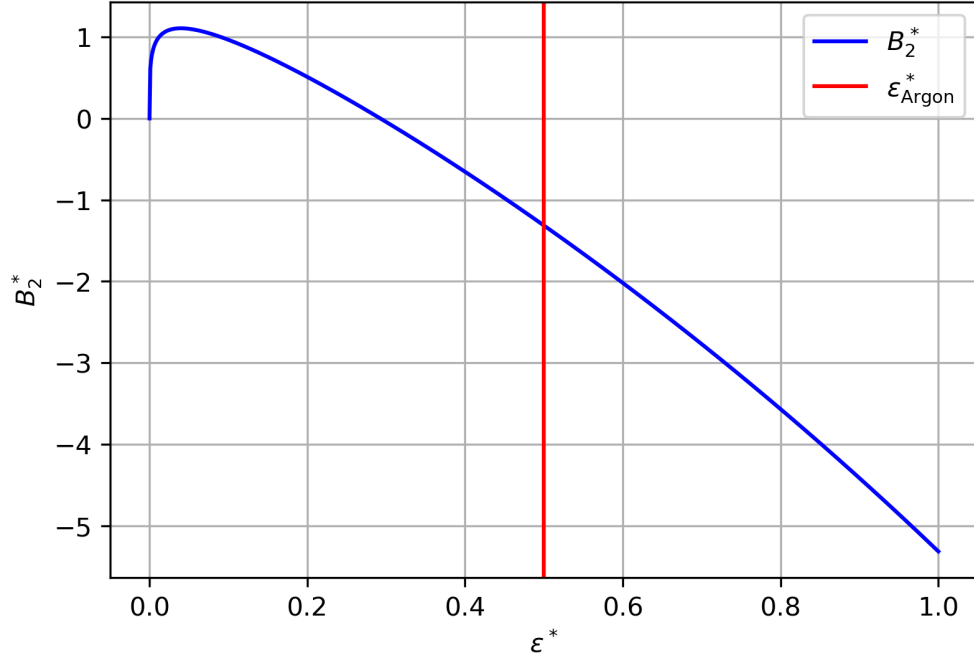


Figure 1: B_2^* for $\epsilon^* \in [0, 1]$.

Part b

Using the Andersen thermostat and the given settings from the sheet, we simulated and evaluated the pressure, temperature and radial distribution function of the system.

The time evolution of the pressure P and temperature T can be found in fig. 2, where one can clearly see that both are fluctuation around a stable mean value, like one would also expect from an equilibrated system.

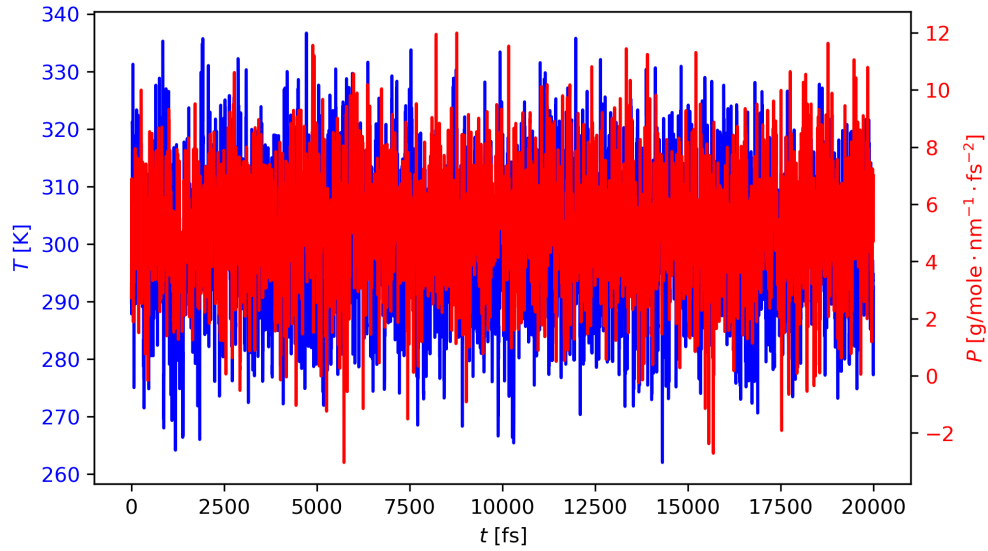


Figure 2: $P(t)$ and $T(t)$ in the production run.

Also the $g(r)$ which can be found in fig. 3 shows the behavior known from earlier sheets.

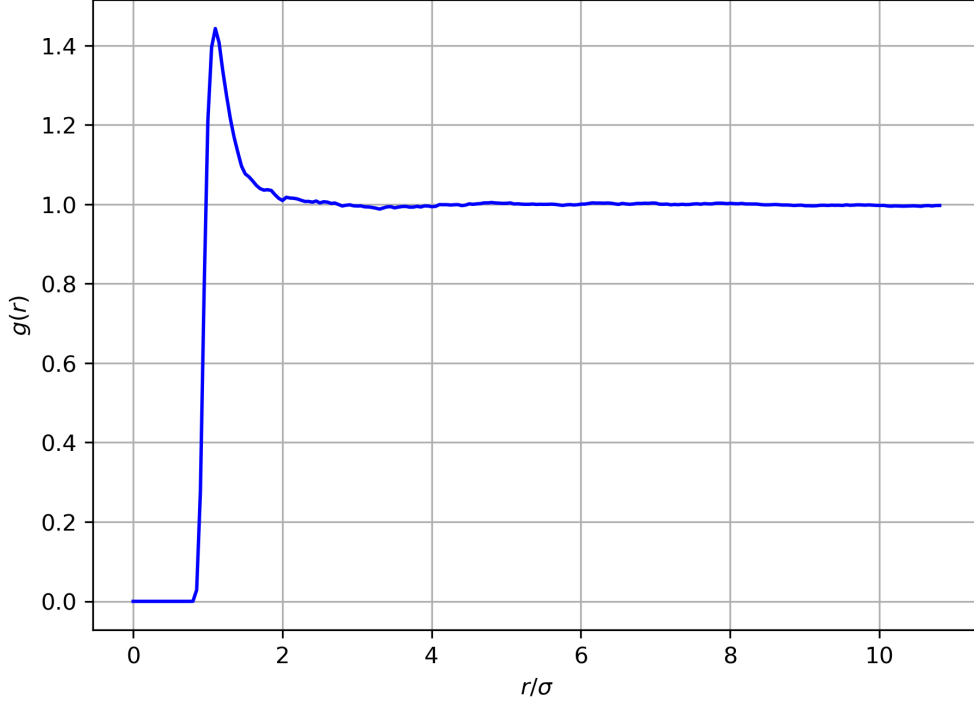


Figure 3: $g(r)$ in the production run.

We then separated the pressures into 5 blocks of equal length and computed the average pressure for each block, resulting in the values

$$\langle P \rangle_1 = 5.219, \quad \langle P \rangle_2 = 5.241, \quad \langle P \rangle_3 = 5.244, \quad \langle P \rangle_4 = 5.253, \quad \langle P \rangle_5 = 5.216,$$

where each value is in the units of $\text{g/mole} \cdot \text{nm}^{-1} \cdot \text{fs}^{-2}$.

Using these values, we calculated the overall mean and the statistical error using the standard deviation divided $\sqrt{5}$, which can all be found in `main.py`. This gave us a value of

$$\langle P \rangle = 5.235 \pm 0.007 \text{ g/mole} \cdot \text{nm}^{-1} \cdot \text{fs}^{-2},$$

which clearly differs from the theoretical expectation of

$$\langle P \rangle_{\text{theo}} = 4.186 \text{ g/mole} \cdot \text{nm}^{-1} \cdot \text{fs}^{-2}$$

which we computed using the $B_2 = B_2^* \cdot \sigma^3$ of our system and plugging it in to the equation of state, eq. (2). This significant deviation could be explained when looking at the equation of state: we used the first order correction (see eq. (2)) to the ideal gas and hereby neglected higher orders, which – already at this low density and thus low contributions of the higher orders – seems to lead to high deviations compared to the general expression (see eq. (1)).

To be honest, this was quite surprising to us since ρ^3 is very small for $\rho = 0.05$, but we checked our code and especially the implementation of the pressure measurement in `update.py` many times and couldn't find an error...

Part c

In the last part, we repeated these steps for 7 densities between $\rho_{\min} = 0.05$ and $\rho_{\max} = 0.25$ and computed the mean pressure $\langle P \rangle$ and the corresponding statistical error by separating into 5 blocks like earlier. The results can be viewed in fig. 4, where they are compared to the ideal gas prediction for the corresponding densities.

The ideal gas behavior shows the expected linearity, while also our measured pressure indicates clearly a linear dependency on ρ , meaning that the pressure of our system doesn't seem to deviate that much of the ideal gas. However, one sees a constant (viewed in the log-log-plot) offset between those two. This clearly indicates that again something is wrong with our scaling – but having wasted enough time in such things in previous sheets and having already spent enough time on this sheet as well, it is pointless to search for this scaling factor any longer.

Looking at our implementation of the pressure measurement in `update.py`, we think that our computations are correct.

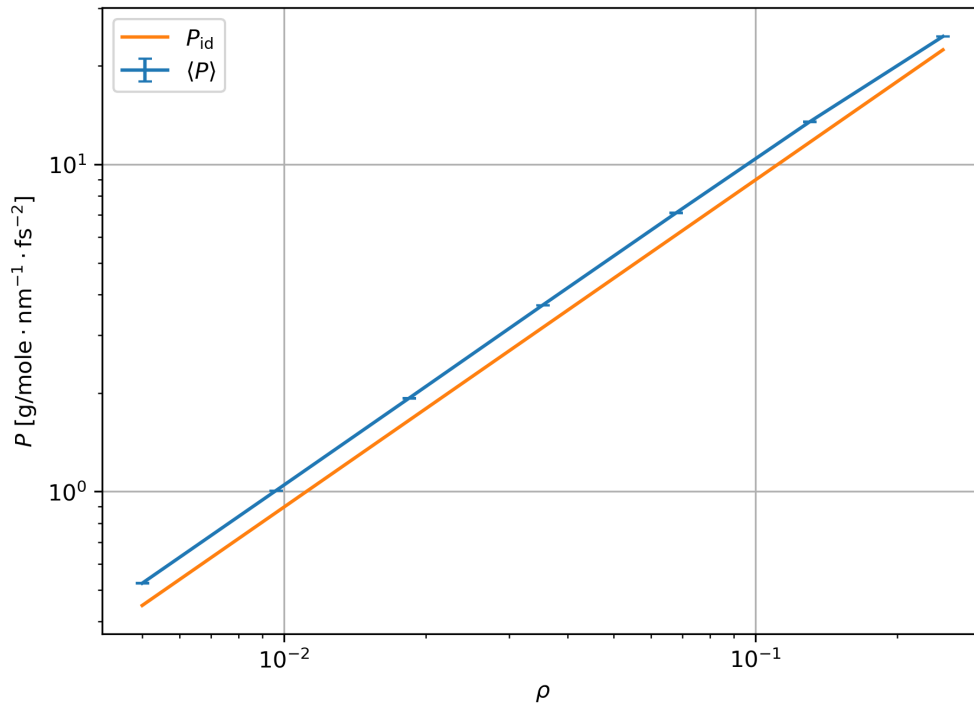


Figure 4: $\langle P \rangle$ and P_{id} for various ρ in a double logarithmic plot.

Given this offset, it is a bit pointless to subtract the ideal gas term $P_{\text{id}} = 1/\beta \cdot \rho$, however we still did it for the sake of completeness. This can be seen in fig. 5.

What we see in fig. 5 however is that for high ρ , the linear behavior alters into something else, which clearly indicates a scaling with $\rho^{N>1}$.

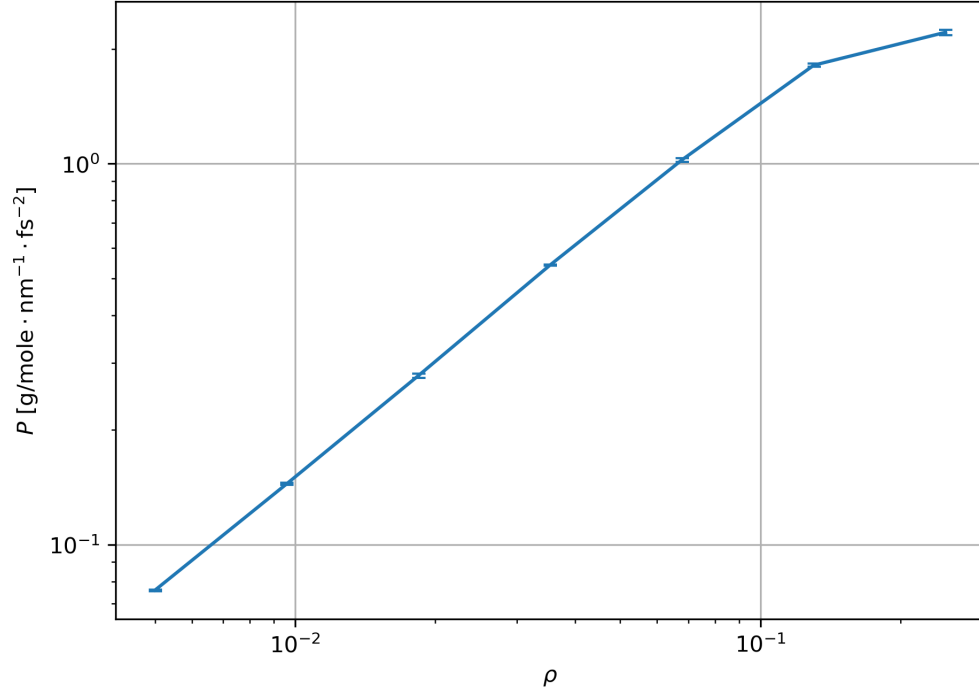


Figure 5: $\langle P \rangle - P_{\text{id}}$ for various ρ in a double logarithmic plot.

Like in the previous tasks, we also analyzed the $g(r)$'s for all densities, which yielded the expected behavior as visualized in fig. 6.

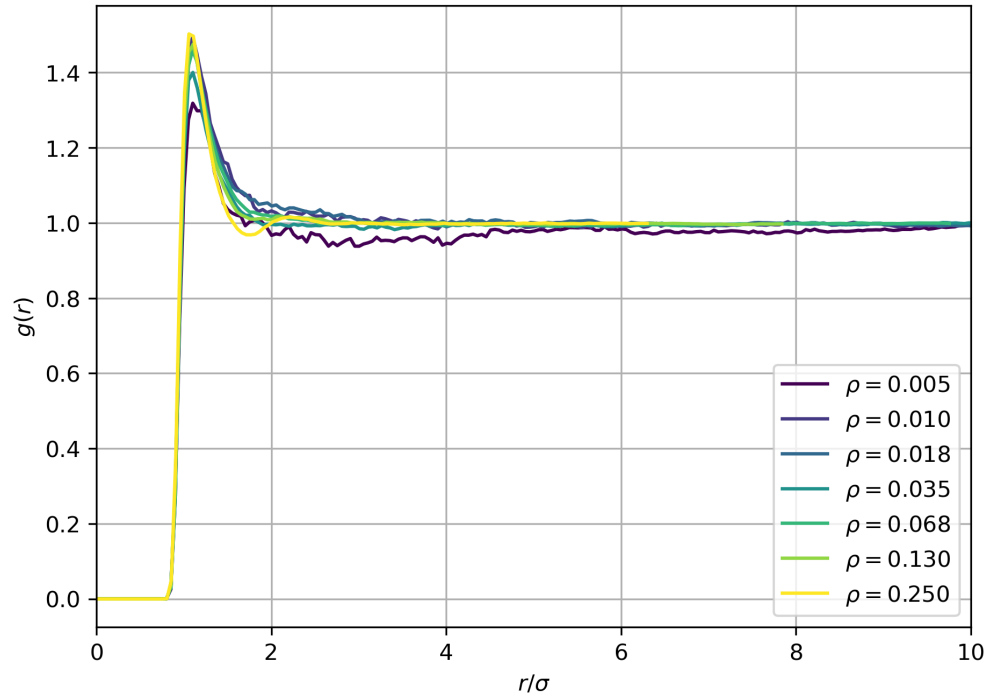


Figure 6: $g(r)$ for various ρ .

In general:

At low densities, particles are on average far apart, so attractive interactions dominate. As discussed in part a, this leads to a negative B_2 , implying that the pressure is reduced compared to the ideal gas: $\langle P \rangle < P_{\text{id}}$.

As the density increases, the effect of B_2 becomes stronger. At sufficiently high densities, higher orders become relevant, and repulsive interactions generally dominates since the particles are very close to each other.

To answer the last question of what happens at $\epsilon^* = 0.1$, we take a look back at fig. 1, where we saw that B_2^* was positive. This means that the deviation we should have observed earlier would be flipped and scaled by $B_2^*(\epsilon^* = 0.1)/B_2^*(\epsilon^* = 0.5)$.